

# SC2006 Software Engineering

## Lecture 22 Supplement

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# Plan for the rest of the semester

- Monday 14th April
  - example past paper questions
- Thursday 17th April
  - non-assessed case study lecture

# Past paper - 2022-2023 Sem 2

Q1

**Identify actors**

# Past paper - 2022-2023 Sem 2

## DISCLAIMER

I have **no special knowledge** of the answers to this past paper, I am talking through **best effort** (and slightly rushed) answers that I've written based on my own knowledge and experience.

I encourage you to **use your own judgement** when revising, and watch the recorded lectures for any clarifications I make regarding the content of the slides

# Past paper - 2022-2023 Sem 2

Q1

Identify actors

User, Company

I'm interpreting the chatbot as the system itself, not an actor!

# Past paper - 2022-2023 Sem 2

Q1

Identify actors

User, Company

**Use cases**

# Past paper - 2022-2023 Sem 2

Q1

Use cases

User

- query the chatbot

- provide product information

- add items to shopping cart

- place orders

- provide order status updates

- process replacement/refund request

# Past paper - 2022-2023 Sem 2

Q1

Use cases

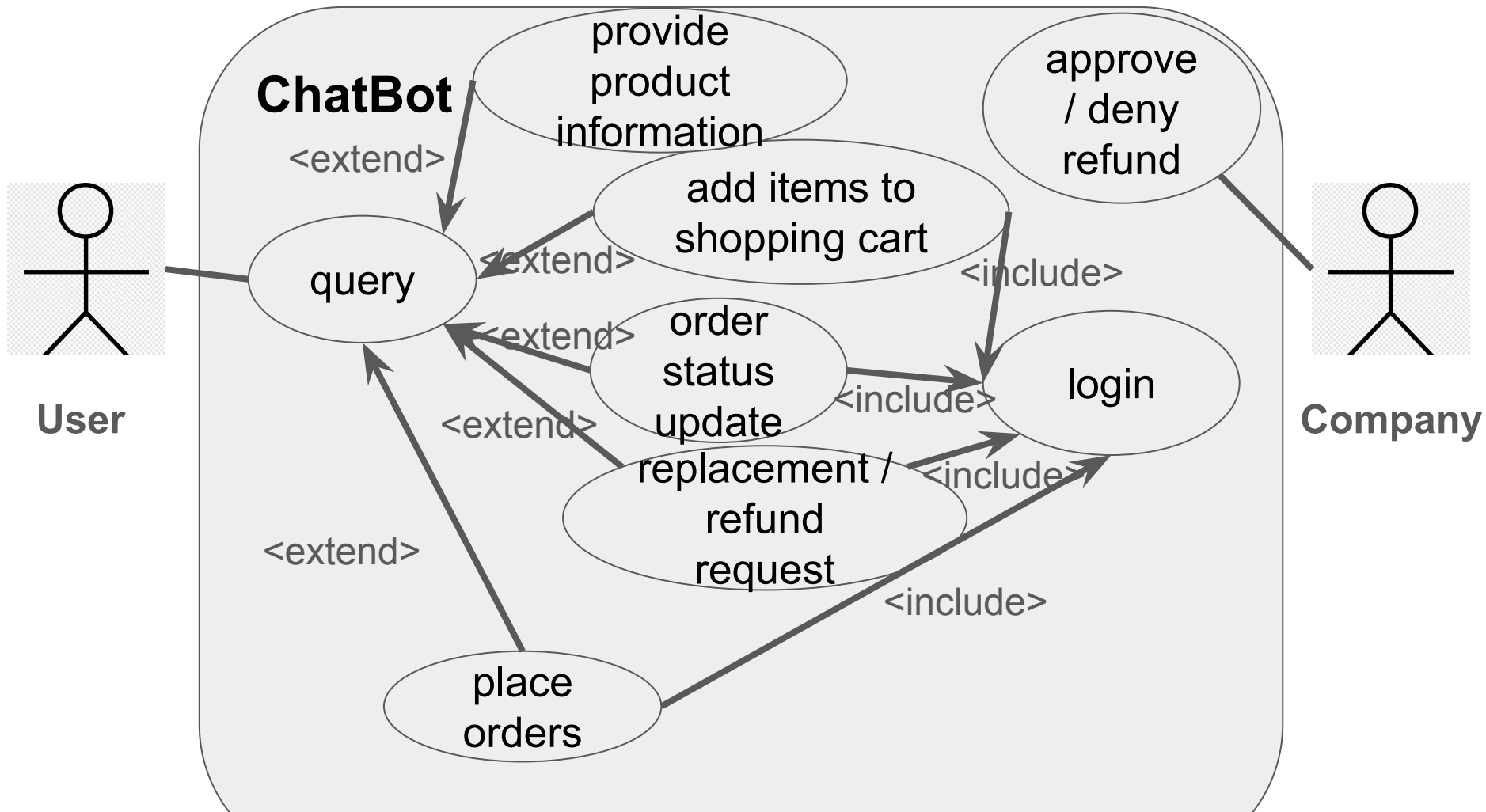
Company

approve or deny refund request



# Past paper - 2022-2023 Sem 2

Q1



# Past paper - 2022-2023 Sem 2

Q1b

Use case description - request replacement

Initiating actor: User

Other actors: Company

Pre-condition:

Request refers to a valid order

Inputs: order id

# Past paper - 2022-2023 Sem 2

## Q1b

1. assert: the order was delivered within the past 7 days
2. the chatbot requests that the user provides item pictures, receipt of purchase, and reason for refund
3. the information is sent to the company
4. if the company accepts the refund, the chatbot displays the replacement procedures to the customer

# Past paper - 2022-2023 Sem 2

Q1b

AF-1: the order is not delivered in the last 7 days

1. the chatbot explains that the refund is not allowed
2. end

AF-2: the user does not provide the requested information correctly

1. the chatbot displays an error message
2. go back to main step 1

AF-4: the company denies the refund

1. the chatbot displays the company's reason for denial
2. end

# Past paper - 2022-2023 Sem 2

Q2a

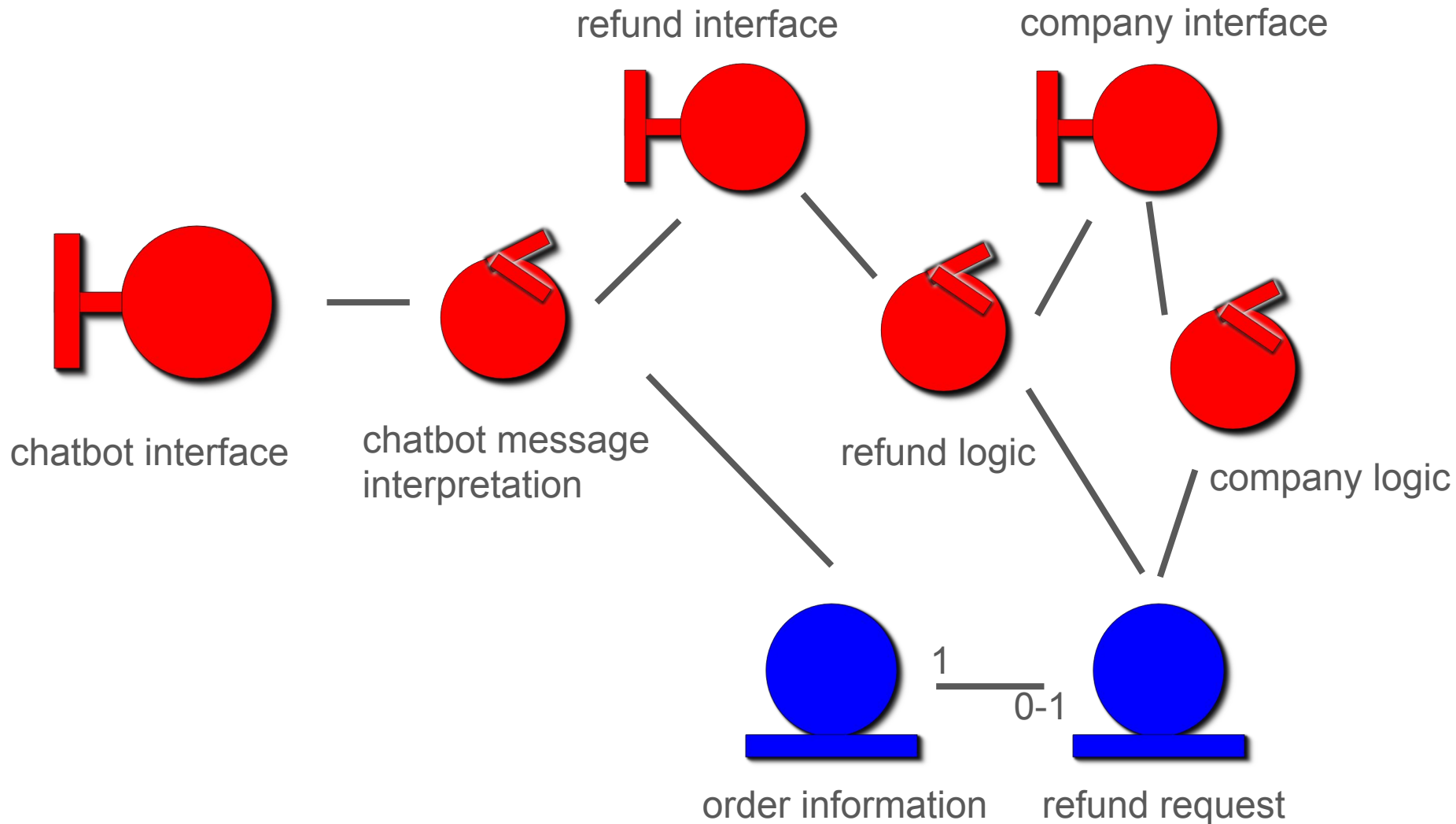
main classes:

boundary: chatbot interface, company interface,  
refund interface(?)

control: chatbot message interpretation, refund logic,  
company logic

entity: order information, refund request

# Past paper - 2022-2023 Sem 2



# Past paper - 2022-2023 Sem 2

Q2b

(I ran out of time to draw this)

roughly - the users messages go through the chatbot interface and are forwarded through the refund interface

checks on the validity of the order require the activation of the order information object

activations proceed as in the use case descriptions

# Past paper - 2022-2023 Sem 2

## Q2c

integration and configuration software process model refers to reusing and composing existing software components to build new systems.

if the existing components don't quite fit the needs of the new system, there is a trade-off to be made between using these components as-is vs spending resources to update / enhance the to give the full functionality of the new system

can be addressed by budgeting time/resources for further extension / adaptation of existing components, or by negotiating with client to adapt requirements to capabilities of existing components



# Past paper - 2022-2023 Sem 2

Q2d

evaluate the impact on user story priorities and points

as a result, possibly reorder or reassign user stories

often, we follow a rule where the tasks of the current sprint are fixed - in this case we could consider ending the sprint early and starting another sprint with appropriate tasks to tackle the issue

talk to the client about reprioritisation?

# Past paper - 2022-2023 Sem 2

Q3a

The two architectures you've seen in detail in the course are layered architectures and MVC

so you can pick one and discuss the other

advantages of layered architecture

simple, clear hierarchy between different components, can be easily nested to describe subsystems

advantages of MVC architecture

describes reactive, dynamic UI more faithfully

allows us to more faithfully model switching between different views

Past paper - 2022-2023 Sem 2

Others (answered verbally, check the lecture recording)